

CHAPTER – XII APPENDIX

12.1 – Source Code

app.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html')

@app.route('/about')
def about():
    return render_template('about.html')

@app.route('/dashboard')
def dashboard():
    return
    render_template('dashboard01.html')

@app.route('/story')
def story():
    return render_template('story.html')

if __name__ == '__main__':
    app.run(debug=True)
```

Index.html

```
<!DOCTYPE html>
<html>
<head>
<title>Cosmetic Insights</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap
.min.css" rel="stylesheet">

<link rel="stylesheet" href="{{ url_for('static',
filename='css/style.css') }}">
</head>
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
<body>

<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
<div class="container">
<a class="navbar-brand" href="/">Cosmetic Insights</a>

<button class="navbar-toggler" data-bs-toggle="collapse" data-bs-
target="#menu">
<span class="navbar-toggler-icon"></span>
</button>

<div class="collapse navbar-collapse" id="menu">
<ul class="navbar-nav ms-auto">

<li class="nav-item">
<a class="nav-link" href="/">Home</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/about">About</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/dashboard">Dashboard</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/story">Story</a>
</li>

</ul>
</div>
</div>
</nav>

<section class="hero">
<div class="container text-center">

<h1 class="display-4 fw-bold">
Cosmetic Insights
</h1>

<p class="lead">
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

Navigating Cosmetics Trends and Consumer Insights with Tableau

```
<a href="/dashboard" class="btn btn-light btn-lg">
Explore Dashboard
</a>
```

```
</div>
</section>
```

```
<section class="container py-5">
```

```
<h2 class="section-title">
Project Highlights
</h2>
```

```
<div class="row">
```

```
<div class="col-md-4">
<div class="card shadow p-4">
<h4>Brand Analysis</h4>
<p>
Identify top cosmetic brands based on sales,
reviews and product popularity.
</p>
</div>
</div>
```

```
<div class="col-md-4">
<div class="card shadow p-4">
<h4>Skin Type Insights</h4>
<p>
Analyze products suitable for Dry, Oily,
Combination and Sensitive skin.
</p>
</div>
</div>
```

```
<div class="col-md-4">
<div class="card shadow p-4">
<h4>Trend Exploration</h4>
<p>
Discover consumer preferences and emerging
beauty trends through Tableau.
</p>
</div>
</div>
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
</p>
</div>
</div>

</div>
</section>
```

```
<div class="footer">
  © 2026 Cosmetic Insights Project
</div>
```

```
</body>
</html>
```

About.html

```
<!DOCTYPE html>
<html>
<head>
```

```
<title>About</title>
```

```
<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap
.min.css" rel="stylesheet">
```

```
<link rel="stylesheet" href="{{ url_for('static',
filename='css/style.css') }}">
```

```
</head>
```

```
<body>
```

```
<!-- Navigation Bar -->
```

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark sticky-top">
```

```
<div class="container">
```

```
<a class="navbar-brand fw-bold" href="/">
```

```
  Cosmetic Insights
```

```
</a>
```

```
<button class="navbar-toggler" type="button"
```

```
  data-bs-toggle="collapse"
```

```
  data-bs-target="#navbarNav">
```

```
  <span class="navbar-toggler-icon"></span>
```

```
</button>
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
<div class="collapse navbar-collapse" id="navbarNav">
  <ul class="navbar-nav ms-auto">

    <li class="nav-item">
      <a class="nav-link" href="/">Home</a>
    </li>

    <li class="nav-item">
      <a class="nav-link" href="/about">About</a>
    </li>

    <li class="nav-item">
      <a class="nav-link"
href="/dashboard">Dashboard</a>
    </li>

    <li class="nav-item">
      <a class="nav-link" href="/story">Story</a>
    </li>

  </ul>
</div>
</nav>
<!-- Project Overview Section -->
<!-- Project Overview Section -->
<section class="container my-5">
  <div class="text-center mb-5">
    <h1 class="fw-bold text-primary">Project Overview</h1>
    <p class="lead">
      Cosmetic Insights: Navigating Cosmetics Trends and
Consumer Insights with Tableau.
      This project analyzes cosmetic product trends, consumer
preferences, brand performance,
      and skin-type suitability using Tableau visualizations
and dashboards.
    </p>
  </div>
</section>

<!-- Team Members Section -->
<section class="container my-5">
  <div class="text-center mb-5">
    <h2 class="fw-bold text-dark">Our Team Members</h2>
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
<p class="text-muted">Meet the talented team behind this project.</p>
</div>
```

```
<div class="row g-4">
```

```
<!-- Team Member 1 -->
<div class="col-lg-3 col-md-6">
  <div class="card shadow border-0 h-100 text-center">
    <div class="card-body">
      <h4 class="fw-bold text-primary">REVURU CHENCHU
NANI</h4>
      <p class="fw-semibold">Project Leader</p>
      <p class="card-text">
        Responsible for project planning,
coordination, and dashboard design.
      </p>
    </div>
  </div>
</div>
```

```
<!-- Team Member 2 -->
<div class="col-lg-3 col-md-6">
  <div class="card shadow border-0 h-100 text-center">
    <div class="card-body">
      <h4 class="fw-bold text-success">KAMASANI
PREMCHANDU</h4>
      <p class="fw-semibold">TEAM MEMBER</p>
      <p class="card-text">
        Performed data cleaning, preprocessing, and
trend analysis.
      </p>
    </div>
  </div>
</div>
```

```
<!-- Team Member 3 -->
<div class="col-lg-3 col-md-6">
  <div class="card shadow border-0 h-100 text-center">
    <div class="card-body">
      <h4 class="fw-bold text-warning">KONGARA
PRAKASH</h4>
      <p class="fw-semibold">TEAM MEMBER</p>
      <p class="card-text">
```


Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

Created interactive dashboards, charts, and story visualizations.

```
        </p>
      </div>
    </div>
  </div>

  <!-- Team Member 4 -->
  <div class="col-lg-3 col-md-6">
    <div class="card shadow border-0 h-100 text-center">
      <div class="card-body">
        <h4 class="fw-bold text-danger">KOTAKONDA
ANIL</h4>
        <p class="fw-semibold">TEAM MEMBER</p>
        <p class="card-text">
          Prepared reports, project documentation, and
presentation materials.
        </p>
      </div>
    </div>
  </div>
</div>
</section>

</body>
</html>
```

Dashboard01.html

```
````html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-
scale=1.0">
 <title>Cosmetic Insights Dashboard</title>

 <!-- Bootstrap CSS -->
 <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstr
ap.min.css" rel="stylesheet">

 <style>
```

# Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

---

```
body{
 background-color:#f8f9fa;
 font-family:'Segoe UI',sans-serif;
}

/* Navbar */
.navbar{
 box-shadow:0 2px 10px rgba(0,0,0,0.15);
}

/* Hero Section */
.hero-section{
 background:linear-gradient(135deg,#ff758c,#ff7eb3);
 color:white;
 text-align:center;
 padding:60px 20px;
}

.hero-section h1{
 font-weight:700;
}

/* Dashboard Container */
.dashboard-section{
 padding:30px 15px;
}

.dashboard-card{
 border:none;
 border-radius:15px;
 overflow:hidden;
}

.tableau-container{
 width:100%;
 overflow:hidden;
}

.tableauPlaceholder{
 width:100%;
}

.tableauViz{
 width:100% !important;
```



# Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

---

```
}

/* Footer */
footer{
 background:#212529;
 color:white;
 text-align:center;
 padding:15px;
 margin-top:30px;
}
</style>
</head>

<body>

<!-- Navbar -->
<nav class="navbar navbar-expand-lg navbar-dark bg-dark sticky-top">
 <div class="container">

  Cosmetic Insights

 <button class="navbar-toggler"
 type="button"
 data-bs-toggle="collapse"
 data-bs-target="#navbarNav">

 </button>

 <div class="collapse navbar-collapse" id="navbarNav">

 <ul class="navbar-nav ms-auto">

 <li class="nav-item">
 Home

 <li class="nav-item">
 About

 </div>
 </div>
</nav>
```

## Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

---

```
<li class="nav-item">

 Dashboard

<li class="nav-item">
 Story

</div>
</div>
</nav>

<!-- Hero Section -->
<section class="hero-section">

 <div class="container">

 <h1 class="display-4">
 Cosmetic Insight Dashboard
 </h1>

 <p class="lead">
 Navigating Cosmetics Trends and Consumer Insights with
 Tableau
 </p>
 </div>
</section>

<!-- Dashboard -->
<section class="dashboard-section">

 <div class="container-fluid">

 <div class="card dashboard-card shadow-lg">

 <div class="card-body">
```

# Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

---

```
<h2 class="text-center text-primary fw-bold mb-3">
 Interactive Tableau Dashboard
</h2>
```

```
<p class="text-center text-muted mb-4">
 Analyze cosmetic brands, product categories,
 skin types, consumer preferences and market
```

trends.

```
</p>
```

```
<div class="tableau-container">
```

```
 <div class='tableauPlaceholder'
 id='viz1781173360792'
 style='position:relative'>
```

```
 <noscript>
```

```

```

```
 <img alt='Cosmetic Insight
```

Dashboard'

```
src='https://public.tableau.com/static/images/RE/REVURUCHENCHUNANI -
NAVIGATINGCOSMETICTRENDSANDCONSUMERINSIGHTSWITHTABLEAU/CosmeticIns
ightDashboard/1_rss.png'
```

```
style='border:none'>
```

```

```

```
</noscript>
```

```
<object class='tableauViz'
 style='display:none;'>
```

```
 <param name='host_url'
```

```
value='https%3A%2F%2Fpublic.tableau.com%2F' />
```

```
 <param name='embed_code_version'
 value='3' />
```

```
 <param name='site_root'
 value='' />
```

```
 <param name='name'
 value='REVURUCHENCHUNANI -
```

# Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

NAVIGATINGCOSMETICTRENDSANDCONSUMERINSIGHTSWITHTABLEAU/CosmeticInsightDashboard' />

```
<param name='tabs'
value='no' />
```

```
<param name='toolbar'
value='yes' />
```

```
<param name='animate_transition'
value='yes' />
```

```
<param name='display_static_image'
value='yes' />
```

```
<param name='display_spinner'
value='yes' />
```

```
<param name='display_overlay'
value='yes' />
```

```
<param name='display_count'
value='yes' />
```

```
<param name='language'
value='en-US' />
```

```
<param name='filter'
value='publish=yes' />
```

```
</object>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

## Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

---

```
<!-- Tableau Script -->
<script type='text/javascript'>

var divElement =
 document.getElementById('viz1781173360792');

var vizElement =
 divElement.getElementsByTagName('object')[0];

function resizeTableau(){

 if(window.innerWidth > 1400){

 vizElement.style.width='100%';
 vizElement.style.height='1000px';
 }
 else if(window.innerWidth > 1200){

 vizElement.style.width='100%';
 vizElement.style.height='900px';
 }
 else if(window.innerWidth > 992){

 vizElement.style.width='100%';
 vizElement.style.height='850px';
 }
 else if(window.innerWidth > 768){

 vizElement.style.width='100%';
 vizElement.style.height='700px';
 }
 else{

 vizElement.style.width='100%';
 vizElement.style.height='600px';
 }
}

resizeTableau();
```

## Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

---

```
window.addEventListener(
 'resize',
 resizeTableau
);

var scriptElement =
 document.createElement('script');

scriptElement.src =
 'https://public.tableau.com/javascripts/api/viz_v1.js';

vizElement.parentNode.insertBefore(
 scriptElement,
 vizElement
);
</script>

<!-- Footer -->
<footer>
 <p class="mb-0">
 © 2026 Cosmetic Insights |
 Tableau Analytics Project
 </p>
</footer>

<!-- Bootstrap JS -->
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap
.bundle.min.js"></script>
</body>
</html>
...
```

# Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

---

## Story.html

```
```html
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">

  <title>Cosmetic Insights Story</title>

  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstr
ap.min.css" rel="stylesheet">

  <style>

    body{
      background-color:#f8f9fa;
      font-family:'Segoe UI',sans-serif;
    }

    /* Navbar */
    .navbar{
      box-shadow:0 2px 10px rgba(0,0,0,0.15);
    }

    /* Story Banner */
    .story-banner{
      background:linear-gradient(135deg,#6f42c1,#8e44ad);
      color:white;
      text-align:center;
      padding:70px 20px;
    }

    .story-banner h1{
      font-weight:700;
    }

    /* Story Card */
    .story-card{
      max-width:1200px;
```


Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
        margin:auto;
        border:none;
        border-radius:20px;
        overflow:hidden;
    }

    .tableauPlaceholder{
        width:100%;
        margin:auto;
    }

    .tableauViz{
        width:100% !important;
    }

    footer{
        background:#212529;
        color:white;
        text-align:center;
        padding:15px;
        margin-top:40px;
    }

</style>

</head>

<body>

<!-- Navbar -->

<nav class="navbar navbar-expand-lg navbar-dark bg-dark sticky-top">

    <div class="container">

        <a class="navbar-brand fw-bold" href="/">
            🔒 Cosmetic Insights
        </a>

        <button class="navbar-toggler"
            type="button"
            data-bs-toggle="collapse"
            data-bs-target="#navbarNav">
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
<span class="navbar-toggler-icon"></span>

</button>

<div class="collapse navbar-collapse" id="navbarNav">

  <ul class="navbar-nav ms-auto">

    <li class="nav-item">
      <a class="nav-link" href="/">Home</a>
    </li>

    <li class="nav-item">
      <a class="nav-link" href="/about">About</a>
    </li>

    <li class="nav-item">
      <a class="nav-link"
href="/dashboard">Dashboard</a>
    </li>

    <li class="nav-item">
      <a class="nav-link active"
href="/story">Story</a>
    </li>

  </ul>

</div>

</div>

</nav>

<!-- Story Banner -->
<section class="story-banner">

  <div class="container">

    <h1 class="display-4">
      Cosmetic Insights Story
    </h1>
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
<p class="lead">
  Data Driven Beauty Industry Analysis
</p>

</div>

</section>

<!-- Story Section -->
<section class="py-5">
  <div class="container">
    <div class="row justify-content-center">
      <div class="col-lg-11">
        <div class="card story-card shadow-lg">
          <div class="card-body p-4">
            <h2 class="text-center text-primary fw-bold mb-3">
              Interactive Tableau Story
            </h2>
            <p class="text-center text-muted mb-4">
              Discover consumer preferences, product
              categories, skin type analysis, pricing insights,
              and brand trends through an interactive Tableau Story.
            </p>
            <!-- Tableau Story Embed -->

            <div class='tableauPlaceholder'
              id='viz1781173500526'
              style='position:relative'>

              <noscript>
                <a href='#'>
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
<img alt='Cosmetics Story'  
  
src='https://public.tableau.com/static/images/RE/REVURUCHENCHUNANI-  
NAVIGATINGCOSMETICTRENDSANDCONSUMERINSIGHTSWITHTABLEAU-  
Story/Story1/1_rss.png'  
  
style='border:none'>  
  
</a>  
</noscript>  
  
<object class='tableauViz'  
style='display:none;'>  
  
  <param name='host_url'  
value='https%3A%2F%2Fpublic.tableau.com%2F' />  
  
  <param name='embed_code_version'  
value='3' />  
  
  <param name='path'  
value='views/REVURUCHENCHUNANI-  
NAVIGATINGCOSMETICTRENDSANDCONSUMERINSIGHTSWITHTABLEAU-  
Story/Story1?:language=en-US&:embed=true&publish=yes' />  
  
  <param name='toolbar'  
value='yes' />  
  
  <param name='animate_transition'  
value='yes' />  
  
  <param name='display_static_image'  
value='yes' />  
  
  <param name='display_spinner'  
value='yes' />  
  
  <param name='display_overlay'  
value='yes' />  
  
  <param name='display_count'  
value='yes' />  
  
  <param name='language'
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
value='en-US' />

</object>

</div>

</div>

</div>

</div>

</div>

</div>

</div>

</section>

<!-- Tableau Script -->

<script type="text/javascript">

    var divElement =
        document.getElementById('viz1781173500526');

    var vizElement =
        divElement.getElementsByTagName('object')[0];

    vizElement.style.width = '100%';
    vizElement.style.height = '950px';

    var scriptElement =
        document.createElement('script');

    scriptElement.src =
        'https://public.tableau.com/javascripts/api/viz_v1.js';

    vizElement.parentNode.insertBefore(
        scriptElement,
        vizElement
    );

</script>
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
<!-- Footer -->
```

```
<footer>
```

```
  <p class="mb-0">
```

```
    © 2026 Cosmetic Insights | Tableau Analytics Project
```

```
  </p>
```

```
</footer>
```

```
<!-- Bootstrap JS -->
```

```
<script
```

```
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap  
.bundle.min.js"></script>
```

```
</body>
```

```
</html>
```

```
...
```

Style.css

```
body{
```

```
  font-family: 'Segoe UI', sans-serif;
```

```
  background:#f8f9fa;
```

```
}
```

```
.navbar{
```

```
  box-shadow:0 2px 10px rgba(0,0,0,0.1);
```

```
}
```

```
.hero{
```

```
  background:linear-gradient(135deg,#ff758c,#ff7eb3);
```

```
  color:white;
```

```
  padding:120px 0;
```

```
}
```

```
.section-title{
```

```
  text-align:center;
```

```
  margin-bottom:40px;
```

```
  font-weight:bold;
```

```
  color:#333;
```

```
}
```

```
.card{
```

```
  border:none;
```

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

```
    transition:0.4s;
}

.card:hover{
    transform:translateY(-10px);
}

.footer{
    background:#212529;
    color:white;
    padding:20px;
    text-align:center;
    margin-top:50px;
}

.dashboard-container{
    padding:50px;
}

.tableau-frame{
    width:100%;
    height:700px;
    border-radius:10px;
    box-shadow:0 0 15px rgba(0,0,0,.2);
}

.about-banner{
    background:#ff758c;
    color:white;
    padding:60px;
    text-align:center;
}

.story-banner{
    background:#6f42c1;
    color:white;
    padding:60px;
    text-align:center;
}
```